

```

cj@LAPTOP-HHJACBEF:~/cmisc125/cpu-scheduler-simulator$ ./schedsim --algorithm=FCFS --input=tests/workloads/quiz3.txt
./schedsim --algorithm=SJF --input=tests/workloads/quiz3.txt
./schedsim --algorithm=STCF --input=tests/workloads/quiz3.txt
./schedsim --algorithm=RR --input=tests/workloads/quiz3.txt
./schedsim --algorithm=MLFQ --input=tests/workloads/quiz3.txt

=== Gantt Chart ===
Each char = 10 time units:
[AAAA][BBBB][CCCC][DDDD][EEEE]
Time: 0 240 420 570 650 780

=== Metrics ===

```

Process	AT	BT	FT	TT	WT	RT
A	0	240	240	240	0	0
B	10	180	420	410	230	230
C	20	150	570	550	400	400
D	25	80	650	625	545	545
E	30	130	780	750	620	620
Average				515	359	359

```

Convoy effect detected: Process B waited 230 time units

```

```
cj@LAPTOP-HHJACBEF:~/cmcs125/cpu-scheduler-simulator$ ./schedsim --algorithm=RR --quantum=20 -
-input=tests/workloads/workload_large.txt
```

=== Gantt Chart ===

Each char = 10 time units:

```
[AAA][111][222][RRR][333][1111][BBB][111][222][SSS][333][1111][2222][RRR][TTT][III][KKKK][BBBB
][DDD][SSS][333][1111][2222][TTTT][KKK][MMM][DDDD][FFF][UUU][WWW][333][1111][2222][GGG][VVV][
XXX][MMM][OOO][FFFF][HHH][WWW][YYY][333][111][2222][III][XXXX][ZZZ][OOO][QQQ][HHHH][JJJ][AAA]
[YYYY][PPPP][RRR][III][KKK][111][2222][3333][111][222][LLL][3333][111][3333][111][222][MMM][BB
BB][DDD][SSSS][UUU][LLL][NNN][CCC][EEE][TTTT][VVV][MMM][OOO][333][111][3333][111][222][PPP][33
3][1111][3333][111][222][QQQ][333][1111][TTT][MMM][NNN][GGG][HHH][AAA][BBB][UUU][VVV]
Time: 0 10 40 60 80 100 140 150 180 200 220 240 280 320 350 370 390 430
470 480 510 540 580 620 660 690 730 770 780 810 820 850 890 930 940
970 980 1010 1020 1060 1070 1110 1120 1150 1170 1210 1220 1260 1270 1300 1320 1360 1370
1390 1430 1470 1480 1490 1510 1530 1570 1610 1630 1640 1660 1700 1720 1760 1780 1800 181
0 1850 1880 1920 1940 1960 1980 1990 2020 2060 2080 2100 2120 2130 2160 2200 2230 2250 227
0 2280 2320 2360 2390 2410 2430 2440 2480 2490 2500 2510 2520 2530 2540 2550 2560 2565
```

=== Metrics ===

Process	AT	BT	FT	TT	WT	RT
A	0	5	5	5	0	0

```
cj@LAPTOP-HHJACBEF:~/cmcs125/cpu-scheduler-simulator$ ./schedsim --algorithm=FCFS --input=test
s/workloads/workload_staggered.txt
```

=== Gantt Chart ===

```
-----
|           A           |           B           | - |           C           |
-----
0                        10                        14  15                        21
```

=== Metrics ===

Process	AT	BT	FT	TT	WT	RT
A	0	10	10	10	0	0
B	3	4	14	11	7	7
C	15	6	21	6	0	0
Average				9	2	2

```
cj@LAPTOP-HHJACBEF:~/cmisc125/cpu-scheduler-simulator$ ./schedsim --algorithm=MLFQ --input=tests/workloads/workload_mlfq.txt
```

=== Gantt Chart ===

Each char = 10 time units:

[AAA][BBB][CCC][AAAA]

Time: 0 10 20 30 115

=== Metrics ===

Process	AT	BT	FT	TT	WT	RT
A	0	100	115	115	15	0
B	10	10	29	19	9	0
C	20	5	25	5	0	0
Average				46	8	0

```
cj@LAPTOP-HHJACBEF:~/cmisc125/cpu-scheduler-simulator$ ./schedsim --compare --input=tests/workloads/quiz3.txt
```

Running comparison of all scheduling algorithms...

Workload: 5 processes

--- Algorithm Comparison ---

Algorithm	Avg TT	Avg WT	Avg RT	Ctx Sw
FCFS	515.00	359.00	359.00	0
SJF	461.00	305.00	305.00	0
STCF	393.00	237.00	15.00	3
RR	630.60	474.60	1.20	716
MLFQ	614.00	458.00	0.00	169

```
cj@LAPTOP-HHJACBEF:~/cmisc125/cpu-scheduler-simulator$ ./schedsim --algorithm=MLFQ --input=tests/workloads/workload_staircase.txt
```

=== Gantt Chart ===

```
-----  
| A | B | C | D | E | A | B | C | D | E |  
-----  
0   2   4   6   8   10  13  16  19  22  25
```

=== Metrics ===

Process	AT	BT	FT	TT	WT	RT
A	0	5	13	13	8	0
B	1	5	16	15	10	1
C	2	5	19	17	12	2
D	3	5	22	19	14	3
E	4	5	25	21	16	4
Average				17	12	2

```
cj@LAPTOP-HHJACBEF:~/cmisc125/cpu-scheduler-simulator$
```

```
cj@LAPTOP-HHJACBEF:~/cmcs125/cpu-scheduler-simulator$ ./schedsim --algorithm=MLFQ --input=tests/workloads/workload_identical_burst.txt
```

=== Gantt Chart ===

```
-----
| A | B | C |   A   |   B   |   C   |   A   |   B   |   C   |
-----
0   2   4   6           10         14         18         22         26         30
```

=== Metrics ===

Process	AT	BT	FT	TT	WT	RT
A	0	10	22	22	12	0
B	0	10	26	26	16	2
C	0	10	30	30	20	4
Average				26	16	2

```
cj@LAPTOP-HHJACBEF:~/cmcs125/cpu-scheduler-simulator$ ./schedsim --algorithm=MLFQ --input=tests/workloads/workload_single.txt
```

=== Gantt Chart ===

```
-----
|                               A                               |
-----
0                               10
```

=== Metrics ===

Process	AT	BT	FT	TT	WT	RT
A	0	10	10	10	0	0
Average				10	0	0

Convoy effect detected: Process B waited 230 time units

=== Gantt Chart ===

Each char = 10 time units:

[AAAA][DDDD][EEEE][CCCC][BBBB]

Time: 0 240 320 450 600 780

=== Metrics ===

Process	AT	BT	FT	TT	WT	RT
A	0	240	240	240	0	0
B	10	180	780	770	590	590
C	20	150	600	580	430	430
D	25	80	320	295	215	215
E	30	130	450	420	290	290
Average				461	305	305

Convoy effect detected: Process B waited 590 time units

=== Gantt Chart ===

Each char = 10 time units:

[AAA][BBB][CCC][DDDD][EEEE][CCCC][BBBB][AAAA]

Time: 0 10 20 30 110 240 380 550 780

=== Metrics ===

Process	AT	BT	FT	TT	WT	RT
---------	----	----	----	----	----	----

=== Gantt Chart ===

Each char = 10 time units:

[AAA][BBB][CCC][DDDD][EEEE][CCCC][BBBB][AAAA]

Time: 0 10 20 30 110 240 380 550 780

=== Metrics ===

Process	AT	BT	FT	TT	WT	RT
A	0	240	780	780	540	0
B	10	180	550	540	360	0
C	20	150	380	360	210	0
D	25	80	105	80	0	0
E	30	130	235	205	75	75
Average				393	237	15

```
Convoy effect detected: Process A waited 540 time units
```

=== Gantt Chart ===

Each char = 10 time units:

[AAA][BBBB][CCC][AAA][CCC][AAA][CCC][AAA][CCC][AAA][CCC][AAA][CCC][AAA][CCC][AAA][CCC][AAA][CC
C][AAA][CCC][AAA][CCC][AAA][BBB][CCC][AAA][BBBB][AAAA]

Time: 0	10	430	440	450	460	470	480	490	500	510	520	530	540	550	560	570	580	59
0	600	610	620	630	640	650	660	670	680	730	780							

```

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./schedsim --algorithm=STCF --input=tests/workloads/quiz3.txt
./schedsim --algorithm=RR --input=tests/workloads/quiz3.txt ...
Average |      |      |      | 630 | 474 | 1

Convoy effect detected: Process A waited 540 time units

=== Gantt Chart ===
Each char = 10 time units:
[AAA][BBB][CCC][EEE][BBB][CCC][EEE][CCC][DDD][BBB][CCC][BBB][AAA][DDD][BBB][DDD][EEE][AAA][DDD
][AAA][CCC][EEE][AAA][EEE][BBB][CCC][EEE][CCC][DDD][BBB][CCC][BBB][AAA][DDD][AAA][EEE][BBB][AA
A][CCC][BBB][CCC][EEE][AAA][CCC][BBB][CCC][EEE][AAA][CCC][BBB][CCC][EEE][AAA][CCC][AAA][BBB][A
AA][CCC][BBB][AAA][BBB][AAAA]
Time: 0 10 20 30 40 50 70 80 90 100 120 130 140 150 170 180 190 200 220
230 240 250 270 280 290 300 320 330 340 350 370 380 390 400 420 430 440 45
0 460 470 480 490 500 510 520 530 540 550 560 570 580 590 600 610 620 630 6
40 650 660 680 700 710 780

=== Metrics ===
Process | AT | BT | FT | TT | WT | RT
-----|-----|-----|-----|-----|-----|-----
A      | 0  | 240 | 780 | 780 | 540 | 0
B      | 10 | 180 | 701 | 691 | 511 | 0
C      | 20 | 150 | 657 | 637 | 487 | 0
D      | 25 | 80  | 412 | 387 | 307 | 0
E      | 30 | 130 | 605 | 575 | 445 | 0
-----|-----|-----|-----|-----|-----|-----
Average |    |     |     | 614 | 458 | 0

Convoy effect detected: Process A waited 540 time units

```